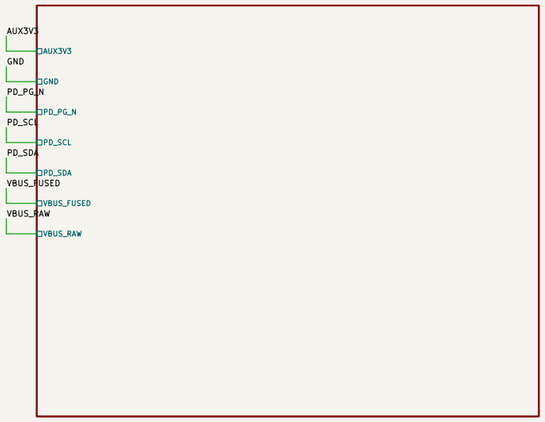
BIG MODULAR / 9 V PD ELECTRICAL PROTOTYPE

CONNECTED REVIEW CANDIDATE — NOT QUALIFIED FOR POWER-UP OR FABRICATION

USB PD and always-on housekeeping



File: 01_input_protection.kicad_sch

Independent hardware gate



File: 02_hardgate.kicad_sch

Independent buck converters



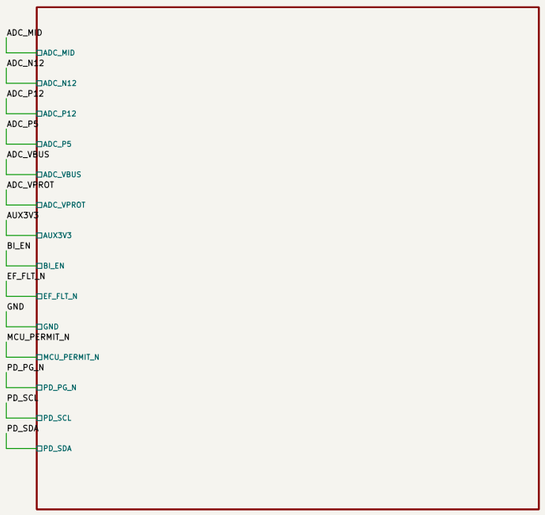
File: 03_bucks.kicad_sch

Direct bipolar rails



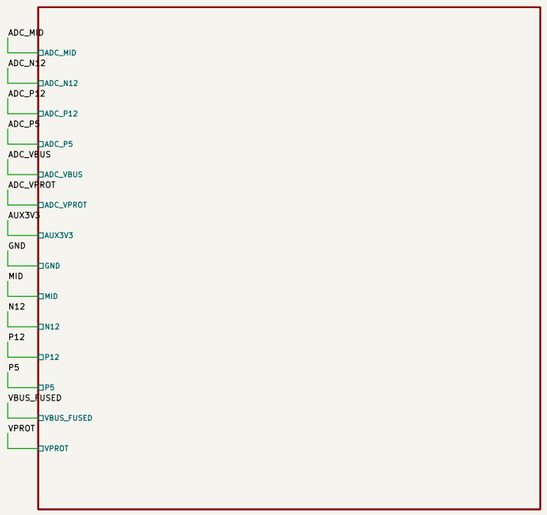
File: 04_bipolar.kicad_sch

Always-on MCU and indicators



File: 05_diagnostics.kicad_sch

Rail sensing and input protection



File: 06_sensing.kicad_sch

Manual output harness



File: 07_output_rails.kicad_sch

READ docs/electrical-prototype.md + firmware/README.md. RUN OPEN: firmware default DENY.
Main stages: TPS54302 MID -> TPS65131 direct +/-12 V; separate TPS54302 +5 V.
Mechanics retained; PCB NOT synchronized or routed. Native parity intentionally incomplete.

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

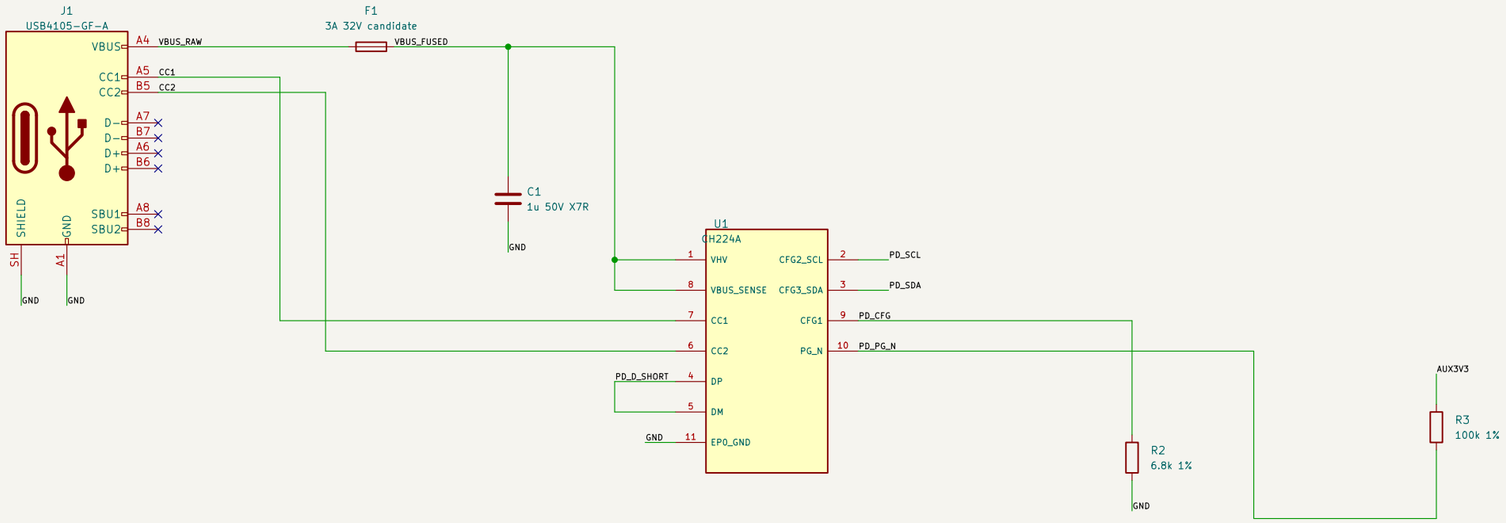
Sheet: /
File: big_modular.kicad_sch

Title: Big Modular — Electrical prototype / hierarchy

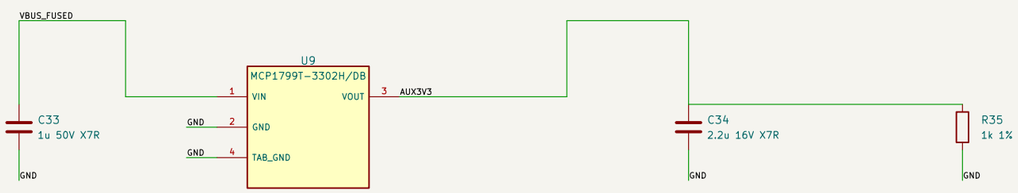
Size: User	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 1/8

01 / USB PD AND ALWAYS-ON HOUSEKEEPING

CH224A: VHV1 + VBUS8 directly fused VBUS; EP0 -> pad11.
WCH section 7.1 contradicts sections 4/6/8 on VBUS rating: vendor confirmation is a POWER-UP BLOCKER.
AUX3V3 operates at negotiation 5 V; NO output-rail operation from fallback. No mains/input = no LEDs.



Direct VBUS feed replaces the old CH224K 1k supply dropper.
I2C 0x22/0x23; status 0x09 bit3 = PD active; current 0x50 x50mA.
PG alone is not proof. Verify WCH silicon/document discrepancy before power.



AUX3V3 AUX3V3
GND GND
PD_PG_N PD_PG_N
PD_SCL PD_SCL
PD_SDA PD_SDA
VBUS_FUSED VBUS_FUSED
VBUS_RAW VBUS_RAW

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

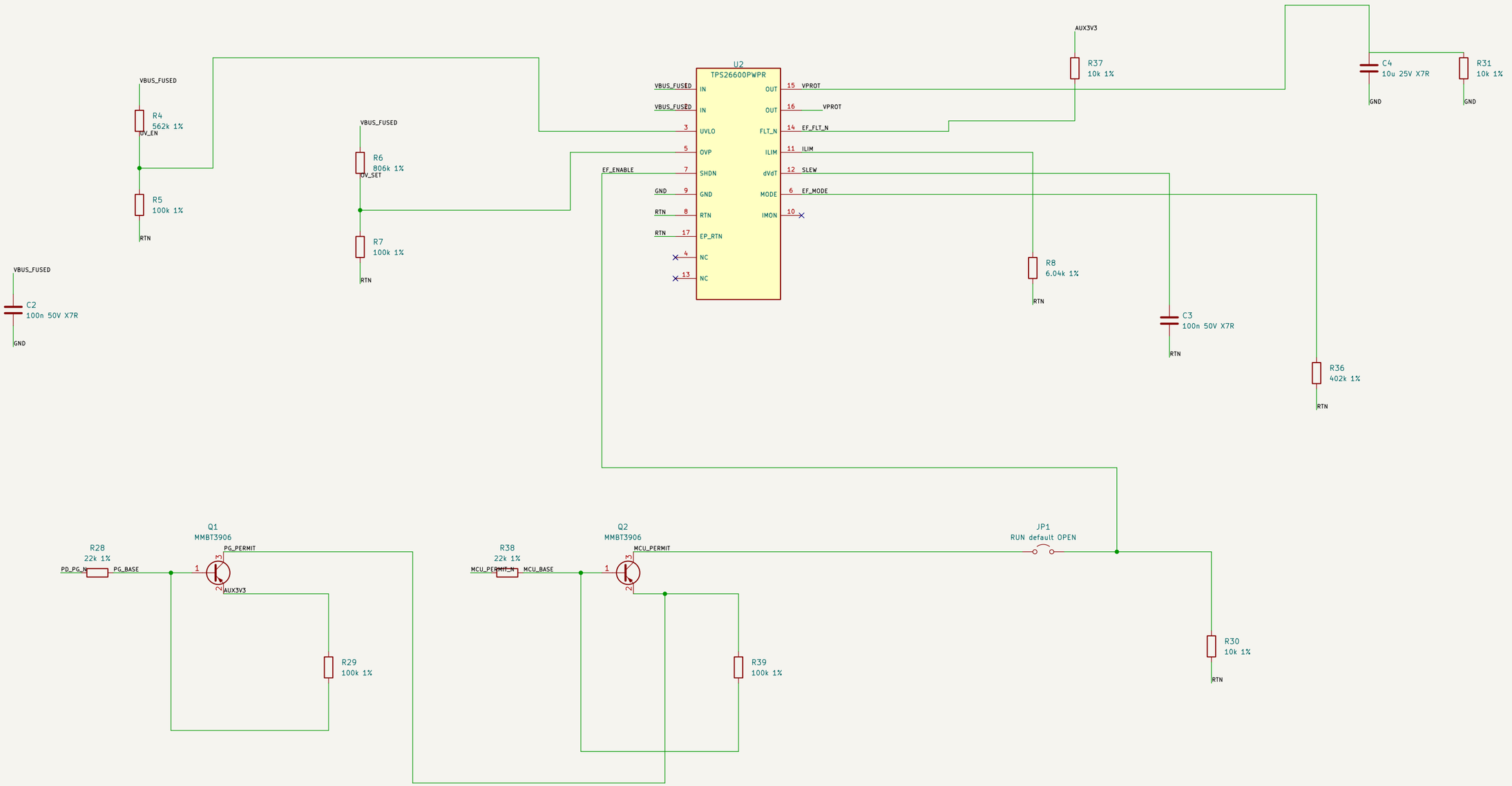
Sheet: /USB PD and always-on housekeeping/
File: 01_input_protection.kicad_sch

Title: Big Modular — USB PD and always-on housekeeping

Size: User	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 2/8

02 / INDEPENDENT HARDWARE GATE

7.878 / 10.781 V nominal hardware window. RTN is NOT GND.
R8 6.04k: 1.987 A nominal, not 3 A. MODE 402k: current limit with thermal latch-off.
JP1 RUN default OPEN: MCU Hi-Z denies via Q2. FLT is aggregate, not per-rail overload proof.



Two PNP permissions in series: PD_PG_N low AND MCU_PERMIT_N low.
RUN must be deliberately closed only after power-up blockers are cleared.

AUX3V3 EF_ENABLE EF_FLT_N GND MCU_PERMIT_N PD_PG_N VBUS_FUSED VPROT

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

Sheet: /Independent hardware gate/
File: 02_hardgate.kicad_sch

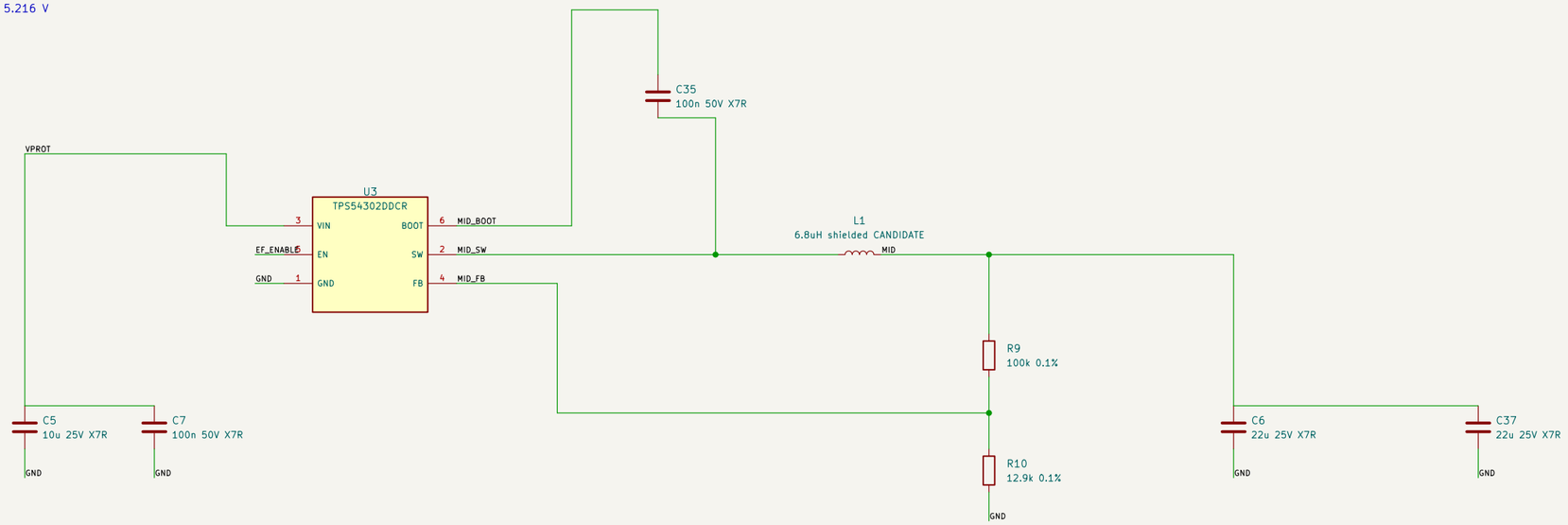
Title: Big Modular — Independent hardware gate

Size: User	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 3/8

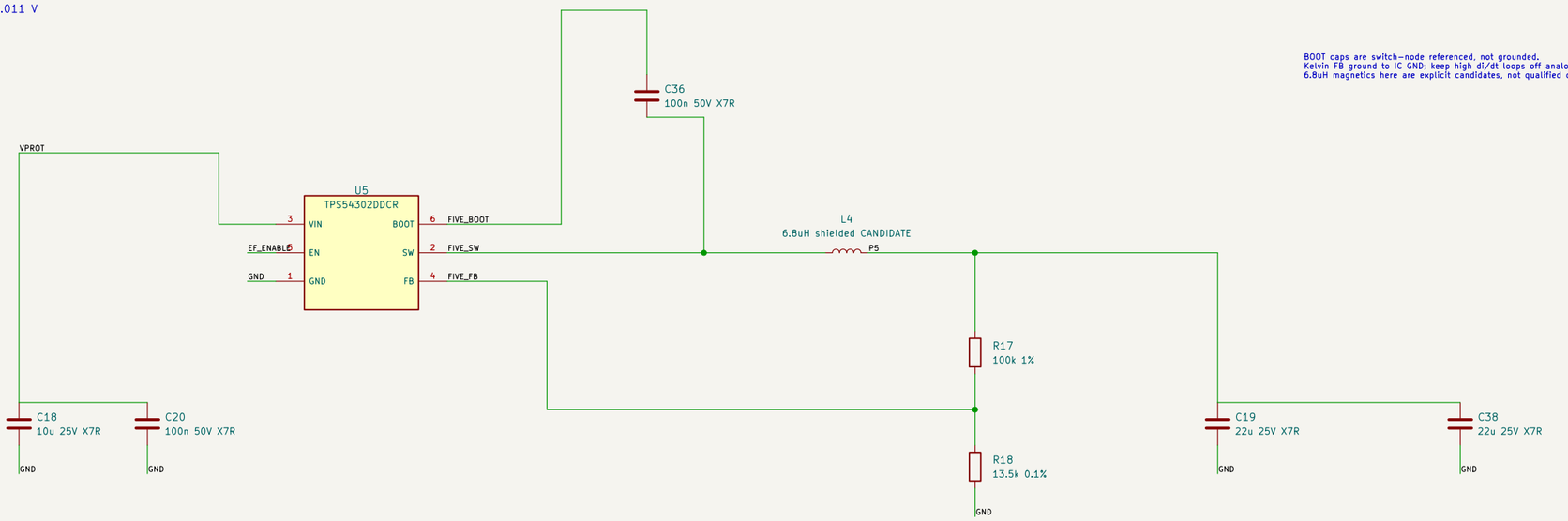
03 / INDEPENDENT BUCK CONVERTERS

U3 MID 5.216 V: upper DC corner 5.357 V BEFORE transients. U4 ceiling 5.5 V.
U5 direct 5.011 V from VPROT, never from positive boost. 100n BOOT-to-SW each.
Effective ceramic C, magnetic Isat/DCR/thermal and loop/transient response remain qualification gates.

U3: MID / 5.216 V



U5: P5 / 5.011 V



BOOT caps are switch-node referenced, not grounded.
Kelvin FB ground to IC GND; keep high dI/dt loops off analog return.
6.8uH magnetics here are explicit candidates, not qualified choices.

EF_ENABLE — EF_ENABLE
GND — GND
MID — MID
P5 — P5
VPROT — VPROT

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

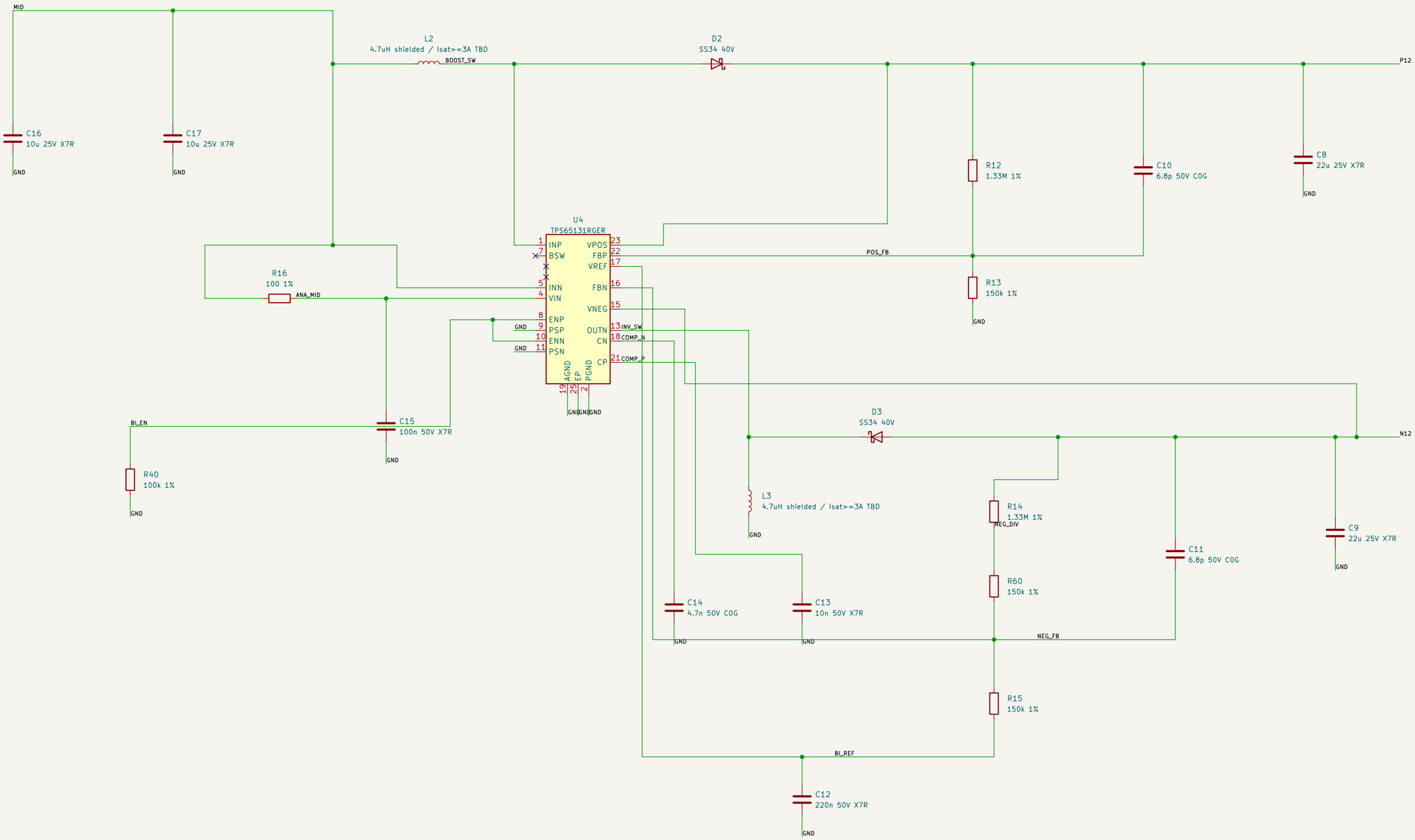
Sheet: /Independent buck converters/
File: 03_bucks.kicad_sch

Title: Big Modular — Independent buck converters

Size: User	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 4/8

04 / DIRECT BIPOLAR RAILS

Nominal +/−11.968 V. FBN returns to VREF, NOT GND; PSP/PSN grounded.
Local output feedback: no remote sense or arbitrary LC added. No rail LDOs.
-800 mA positive is chart interpolation at 25 C, NOT guaranteed continuous or simultaneous +/−800 mA.



Separate INP/INN local ceramic loops; shielded L2/L3.
No arbitrary output LC. Compensation and Cff require loop validation.

ANA_MID ANA_MID
BI_EN BI_EN
GND GND
MID MID
N12 N12
P12 P12

INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE

Big Modular

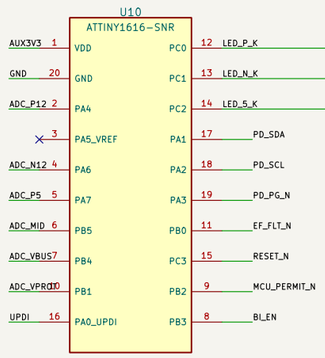
Sheet: /Direct bipolar rails/
File: 04_bipolar.kicad_sch

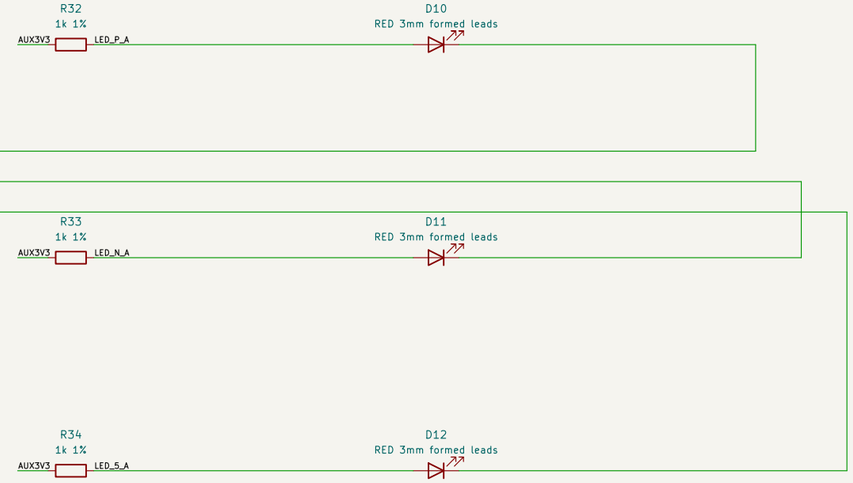
Title: Big Modular — Direct bipolar rails

Size: User	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 5/8

05 / ALWAYS-ON MCU AND INDICATORS

Programming REQUIRED. Default firmware denies the main stage until qualification.
Steady = healthy; 4 Hz affected-rail fault; actual missing PD9 = synchronous 0.5 Hz all LEDs.
RUN open / general FLT / insufficient current are NOT labelled missing PD.

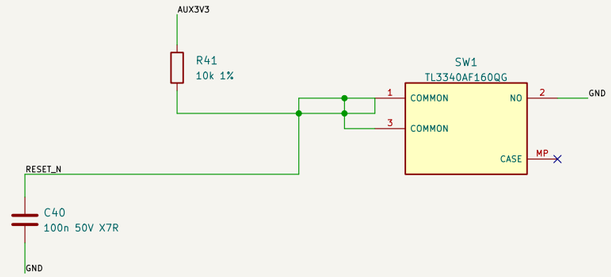




LED supply is ALWAYS AUX3V3, never the monitored rail.
1k resistors R32..R34: Basic 0603; schematic provision only.
D10/D11/D12: standard RED 3mm formed leads, manual JLC exclusion.







PB3 pin8 = independent bipolar enable (default pulldown R40).
PA5 pin3 spare; use internal 2.5 V ADC reference, calibrate before power.

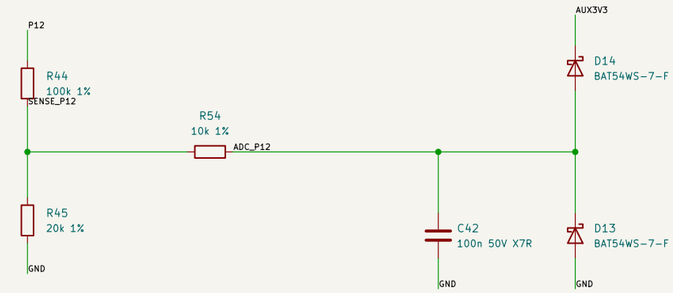
ADC_MID ADC_MID
ADC_N12 ADC_N12
ADC_P12 ADC_P12
ADC_P5 ADC_P5
ADC_VBUS ADC_VBUS
ADC_VPROT ADC_VPROT
AUX3V3 AUX3V3
BI_EN BI_EN
EF_FLT_N EF_FLT_N
GND GND
MCU_PERMIT_N MCU_PERMIT_N
PD_PG_N PD_PG_N
PD_SCL PD_SCL
PD_SDA PD_SDA

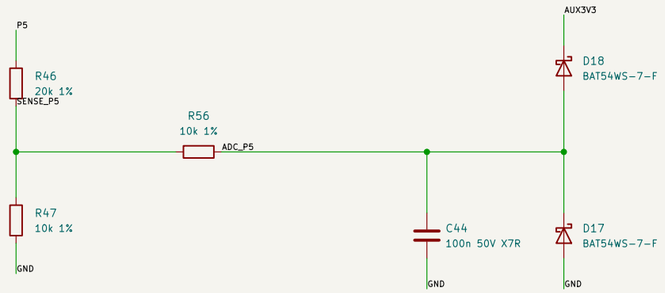
INCOMPLETE / NON-OPERATIONAL / DO NOT FABRICATE
Big Modular
Sheet: /Always-on MCU and indicators/
File: 05_diagnostics.kicad_sch
Title: Big Modular — Always-on MCU and indicators

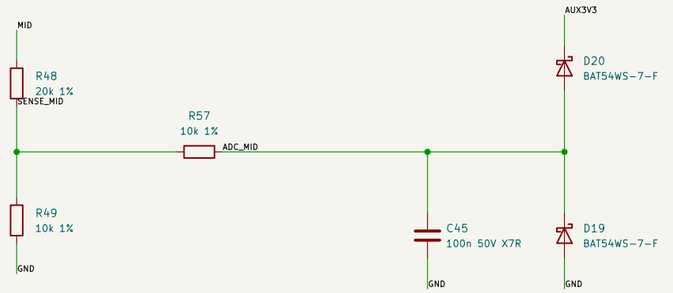
Size: User	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 6/8

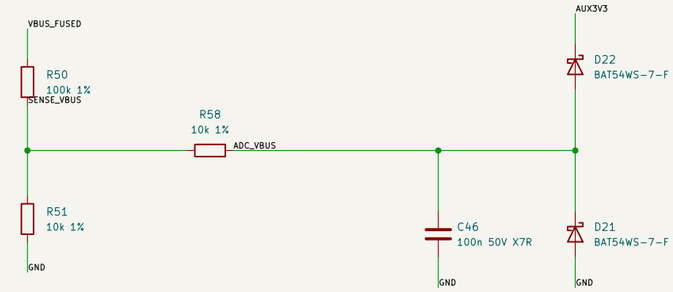
06 / RAIL SENSING AND INPUT PROTECTION

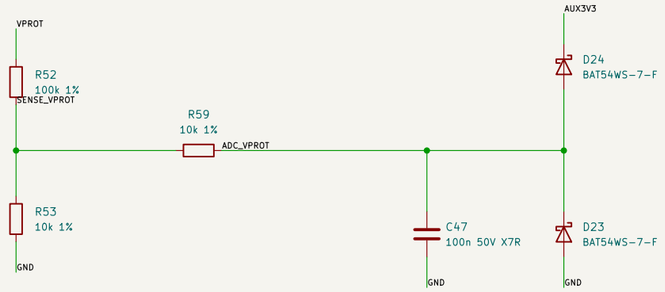
Negative rail uses ground-referenced inverter, NOT AUX-biased divider.
External Schottky clamps + series resistors are provisions, NOT proven powered-off isolation.
OFF-state injection / clamp Vf / residual rail bleed / reference calibration are POWER-UP BLOCKERS.



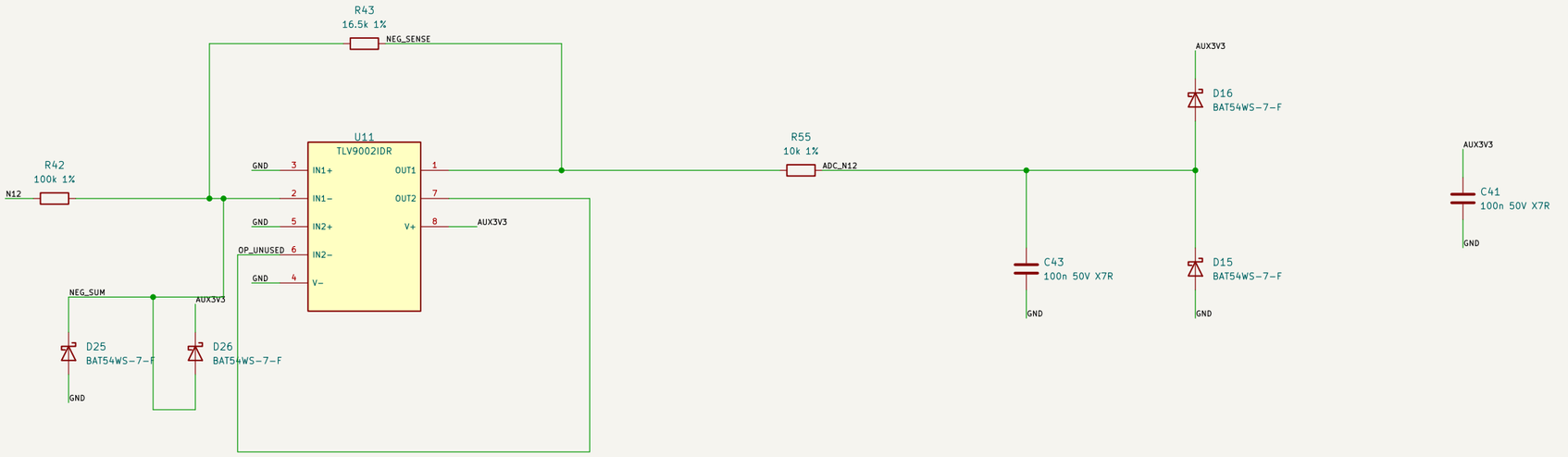








U11A: −VNEG x 16.5k/100k = 1.98 V at −12 V.
U11B unused follower: +Input GND; −Input tied OUT.



ADC_MID ADC_MID
ADC_N12 ADC_N12
ADC_P12 ADC_P12
ADC_P5 ADC_P5
ADC_VBUS ADC_VBUS
ADC_VPROT ADC_VPROT
AUX3V3 AUX3V3
GND GND
MID MID
N12 N12
P12 P12
P5 P5
VBUS_FUSED VBUS_FUSED
VPROT VPROT

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Big Modular

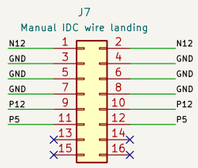
Sheet: /Rail sensing and input protection/
File: 06_sensing.kicad_sch

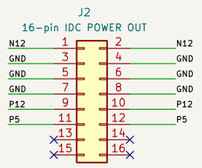
Title: Big Modular — Rail sensing and input protection

Size: User	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 7/8

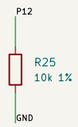
07 / MANUAL OUTPUT HARNESS

J7 board wire landing -> J2 off-board IDC cable, pin-for-pin.
1/2=-12, 3..8=GND, 9/10=+12, 11/12=+5; 13..16 NC.
Never connect another active PSU. Continuity-test red stripe / orientation before use.





Wire landing and IDC are the SAME twelve connected power conductors.
Named continuations retain pin numbers and avoid drawing wires through symbol bodies.
Pins13..16 intentionally not connected. Harness is hand assembled, not JLC placement.























Power flags identify real passive input / converter-output power paths for ERC.
They do not certify operation, current budget or completed PCB connectivity.





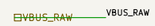












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Big Modular

Sheet: /Manual output harness/
File: 07_output_rails.kicad_sch

Title: Big Modular — Manual output harness

Size: User	Date:	**Rev: Hierarchy WIP**
KiCad E.D.A. 10.0.6		Id: 8/8